

Review



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Author for correspondence:

Jasper Brener

e-mail: jasper.brener@stonybrook.edu

Towards a psychophysics of interoceptive processes: the measurement of heartbeat detection

Jasper Brener¹ and Christopher Ring²

¹Department of Psychology, Stony Brook University, Stony Brook, NY 11790, USA

²School of Sport, Exercise and Rehabilitation Sciences, University of Birmingham, Edgbaston, Birmingham B15 2TT, UK

It is difficult to collect objective evidence of interoception. Unlike exteroception, the effective stimuli for interoception are often unknown, and even when identifiable, they are difficult to control experimentally. Furthermore, direct stimulation of the interoceptors is seldom appropriate in human experimentation. Hence, non-invasive behavioural measures of accuracy in heartbeat detection have frequently been adopted to index interoceptive sensitivity. However, there has been little standardization and the two most popular methods for assessing heartbeat detection, heartbeat tracking and two alternative forced choice methods, appear to be biased and of questionable validity. These issues do not arise with other methods that are based on classical psychophysics and that enable subjects to indicate when during the cardiac cycle their heartbeat sensations occur. Not only are these classical methods highly reliable, but they also provide continuous unbiased measures of the temporal locations of heartbeat sensations and the precision with which these sensations are detected.

This article is part of the themed issue 'Interoception beyond homeostasis: affect, cognition and mental health'.

1. Introduction

Over the past 50 years, research on interoception has grown exponentially from 17 citations of 'interocept*' in 1965 to 8440 in 2015 (Web of Science). The early Russian work was summarized in Razran's landmark article [1] on the subject and demonstrated, *inter alia*, that interoception participated fully in the intersensory processes of Pavlovian conditioning. Through the conventional procedure of presenting stimuli contingently, interoceptive stimuli can acquire the capacity to signal other interoceptive stimuli (interointeroceptive conditioning) or exteroceptive stimuli (interoexteroceptive conditioning) and exteroceptive stimuli can acquire the capacity to signal interoceptive stimuli (exterointeroceptive conditioning). Such conditioning provides routes through which interoception can participate in the regulation of everyday behaviours [2], and in recent years, research has examined its role in a broad range of affective and cognitive processes as well as in personality and psychological disorders [3].

The James–Lange theory of emotion, which attributed a causal role to interoception in generating emotion provided a model for these research developments. Tests of this theory stimulated a line of psychophysiological and neuroscience research that has gathered momentum over the past half century [4,5]. As the motivational and guidance functions of interoception attracted more attention, research broadened to explore its influences on such cognitive processes as implicit and explicit memory [6,7], decision-making [8] and auditory processing [9]. Individual differences in sensitivity to internal stimuli have also been investigated as potential sources of variation in the sense of self [10] and such psychosocial processes as imitation [11] and empathy [12]. They have also been found to be associated with some psychological disorders [13,14] but not others [15,16]. These research developments have been accompanied by detailed neurophysiological models that describe the pathways and processes

by which interoception may achieve its behavioural and psychological effects [5,17–22].

Despite this substantial interest in interoception, standard methods for its investigation have not been adopted. In this regard, the study of interoception has diverged from the history of exteroceptive psychophysics which is grounded in standard methodologies that have provided the foundations of experimental psychology. Fechner's classical methods of adjustment, constant stimuli and limits [23] and Bekey's 'staircase' method [24] enabled thresholds to be precisely and reliably determined for different modalities and individuals. Stevens' power law [25], which depended on the methods of magnitude estimation and production, described our differential sensitivity to features of the physical and social environments.

Recognizing the importance of standard methods in interoceptive research, Garfinkel *et al.* [26] have recently attempted to rectify the 'inconsistency in how interoception is defined and quantified' by proposing a three-dimensional scheme that distinguishes between the measurement of (i) interoceptive accuracy, (ii) interoceptive awareness, and (iii) interoceptive sensibility. They define 'interoceptive accuracy' in terms of the precision and reliability with which individuals detect internal events such as heartbeats or gastric contractions, whereas 'interoceptive awareness' denotes the accuracy with which subjects assess their own precision in detecting such events. Both 'interoceptive accuracy' and 'interoceptive awareness' may be assessed by objective performance measures. In contrast, 'interoceptive sensibility', which refers to the extent to which individuals attend to their internal states, is assessed using self-reports.

This paper is concerned exclusively with measuring the accuracy of heartbeat detection. The methods employed to make these measurements yield most of the data on which current psychophysiological research on interoception relies. They are different from standard psychophysical methods in several respects: notably, the effective dimensions of interoceptive stimuli have not been precisely defined and are not directly controlled by the experimenter.

More conventional psychophysical methods involving direct stimulation of the viscera were employed in the early Russian research on interoception to map the sensitivities of interoceptive domains [27]. This approach was recently adopted by Khalsa *et al.* [28] in a comprehensive examination of the interoceptive effects of sympathetic stimulation. Varying doses of isoproterenol, a non-specific beta-adrenergic agonist, were administered by bolus injections and the responses of subjects were tracked using psychophysiological measures. Dose-dependent effects were found on cardiorespiratory activity and on cardiorespiratory sensations as measured by magnitude production [25]. As dose increased, so too did feelings of physical anxiety and the specificity with which heartbeat sensations were localized in the chest.

Using the same methods on a patient who had nearly complete bilateral lesions of the insula and cingulate cortices, Khalsa *et al.* [19] raised questions about standard views of the receptors responsible for interoceptive sensations and the central structures necessary for viscerosensation and emotion. This patient exhibited 'dose-dependent changes in interoceptive awareness that were comparable to healthy control participants, albeit somewhat delayed in time'. Accordingly, Khalsa and co-workers suggest 'a comprehensive redefinition of interoception involving "afferent information that arises

from anywhere and everywhere within the body" [2], including through the skin via pathways that are usually considered to support exteroception'. Because it is challenging, if not impossible, to identify the pathways through which individuals sense their visceral activities, this redefinition seems necessary.

Direct stimulation methods of the sort used by Khalsa [28] are too intrusive to be used in most experiments concerned with the influence of interoception on emotion, cognition and/or clinical conditions. In such experiments, probes of interoceptive sensitivity should minimize influences on the subjects' affective and cognitive states and should not tire or tax them. These considerations have led to the wide adoption of non-invasive behavioural methods for assessing individual differences in sensitivity to heartbeat sensations. Such methods are minimally intrusive and the best of them provide convincing, objective, quantitative measurements of interoceptive sensitivity. However, as discussed in §§2 and 3, the two methods that have dominated the field, heartbeat tracking and two alternative forced choice (2AFC) discrimination tasks, are subject to substantial criticism.

2. Heartbeat-tracking tasks

Heartbeat tracking encompasses a suite of quick and easy methods designed to measure cardioception by requiring participants to report the number of heartbeats [29,30], tap on each heartbeat [31–34] or adjust the rate of brief exteroceptive stimuli to match their heart rate [35,36].

The heartbeat-counting task introduced by Dale & Anderson [29] and popularized by Schandry [30] asks participants to detect heartbeats during short epochs. The onset and offset of each period (25/35/45 s) is signalled by a brief tone, after which the subject is 'requested to report the counted or estimated number of heartbeats' [30, p. 484]. The absolute difference between the number of actual and reported heartbeats is divided by the number of actual heartbeats either to generate an error score, or subtracted from one to generate a perception score. Perfect detection yields an error score of zero (and a perception score of one), whereas under-reporting or over-reporting the actual number of heartbeats (by up to 100%) yields an error score of greater than zero (and a perception score of less than unity). Schandry's [30] grand mean error score of 0.26 indicated that counted heart rate (59 bpm) underestimated actual heart rate (80 bpm) by 26%. Other studies find similar scores. Schandry & Specht's [37] participants underestimated their heart rates by 36% when resting, 32% before public speaking and 23% after exercise. Ring *et al.*'s [38] participants underestimated their heart rates by 37% when sitting, 42% when standing and 20% after exercise. It seems implausible that normal adults would make counting errors of this magnitude and the substantial under-reporting of the number of heartbeats implies that participants did not reliably detect heartbeat sensations.

Methods have not been devised for determining whether the reported count is based on heartbeat sensations experienced during the task or is an estimate of the number of heartbeats based on the participant's previously acquired beliefs. As noted above, Schandry explicitly told participants to *estimate* the number of heartbeats. Accordingly, participants with knowledge about heart rate may generate accurate counting scores without detecting any heartbeat sensations.

There is convincing evidence that providing accurate information about heart rate via concurrent feedback improves subsequent heartbeat-counting scores [38–40]. That this improvement is also found with cardiac non-contingent feedback indicates that improved counting scores are attributable to improved knowledge rather than improved sensitivity to heartbeats [34,38]. Windmann *et al.*'s [41] research provides additional evidence that counting scores have little connection with heartbeat sensations: changes in heart rate elicited by a cardiac pacemaker were not accompanied by changes in the number of counted heartbeats. Their participants reported heart rates of 52, 54 and 59 bpm when actual heart rates were paced at 61, 76 and 109 bpm, yielding error scores of 0.15, 0.29 and 0.46, respectively. These substantial discrepancies imply that, rather than counting heartbeat sensations, participants based their counts on beliefs about heart rate.

Further evidence against the validity of the counting task comes from experiments on the effects of body posture on counting performance. Studies consistently report that heartbeat-counting scores are better when supine than standing [37,38,40,42,43]. Because stroke volume is augmented in the supine posture, this effect was initially attributed to increased cardiac stimulus intensity. However, later, the increased accuracy was shown to be owing to a reduction in heart rate when moving from the standing to the supine posture that reduced the difference between actual and counted heart rates. For example, Ring & Brener [40] found that counted heart rates did not differ between supine (50 bpm) and standing (48 bpm) despite large differences in actual heart rates between supine (70 bpm) and standing (86 bpm). This latter explanation is supported by the results of five experiments showing that heartbeat detection accuracy is not influenced by passive body tilt [44].

In tapping tasks, e.g. [33], instead of counting, subjects tap or press a key on each heartbeat. Like counting tasks, they may be criticized because individuals, regardless of their cardioceptive sensitivity, can generate accurate scores by tapping (counting) at frequencies that approximate their heart rates [45]. Tapping does, however, provide a means, not available in the covert counting method, of verifying that subjects are detecting heartbeats rather than responding at the rate they believe their hearts to be beating. In particular, the distribution of latencies between R-waves and taps can be analysed for preferences that are time-locked to the R-wave. For example, if a subject presses the button more frequently at around $R + 250$ ms than at other latencies, this would indicate that he/she used heartbeat sensations as signals for tapping. While the tapping method has been used by several investigators [31,32,34,36,46,47], results have been unconvincing.

For example, Flynn & Clemens [48] instructed participants to press a key in synchrony with their heartbeats or sounds. Based on its response latency, each key press was assigned to one of six 100 ms bins (1–100, 101–200, 201–300, 301–400, 401–500, 501–600 ms) following the R-wave or sound. Responses to heartbeats were evenly distributed across the six latency bins indicating that participants could not reliably detect heartbeats. However, this failure could not be attributed to general perceptual limitations because they exhibited non-random distributions in the auditory tapping task. Because heartbeat detection may be evidenced by other procedures, it was inferred that

the demands of button pressing interfered with cardioception by competing for resources with heartbeat detection. Thus, the available evidence indicates that heartbeat-tapping tasks have little to recommend them as methods for assessing cardioception.

It was noted above that in counting and tapping tasks, participants may base their responses on an assumed or believed heartbeat frequency rather than on heartbeat detection. Heart rate estimation tasks generate data on a participant's knowledge about heart rate that might guide such guessing. In these tasks, participants are explicitly asked to estimate their current heart rate rather than to process current heartbeat sensations. Essau & Jamieson [49] subtracted estimated from actual heart rates and found evidence for slight underestimation (–0.8 bpm) to modest overestimation (+5.4 bpm), indicating that participants possess accurate knowledge about resting heart rate. Using magnitude estimation, Pennebaker & Hoover [50] required participants to estimate their heart rates on a scale of 1 (*slow*) to 50 (*normal*) to 100 (*fast*), during various tasks (e.g. meditation, reading, lifting). Overall, heart rate estimates correlated positively and moderately-to-highly with actual heart rates ($r_s = 0.36$ – 0.40 , $p_s < 0.05$). Importantly, this association was independent of any ability to detect heartbeat sensations. Data generated by these estimation tasks suggest that participants are knowledgeable about heart rate. The implication is that tracking (counting/tapping) tasks can be solved by a combination of accurate knowledge and inaccurate interoception.

3. Two alternative forced choice tasks: signal detection methods

A popular method for assessing cardioception that logically requires heartbeat detection for good performance was developed by Whitehead *et al.* [51]. In their 2AFC task, participants were presented with a series of cardiac-contingent light flashes that on half the trials were delayed after the R-wave by 128 ms (S+ trials) and on the other half by 384 ms (S– trials). The S+ and S– designations were based on estimates of when the pressure pulse wave generated by ventricular contraction would stimulate mechanoreceptors in and near the heart to produce heartbeat sensations. Heartbeat sensations should be simultaneous with S+ stimuli and non-simultaneous with S– stimuli. Based on signal detection theory, a decision that heartbeats and light flashes were simultaneous was classified as a *hit* on S+ trials and a *false alarm* on S– trials. Whitehead's [51] grand median d' of 0.33 (0.05–1.56) indicates that most participants struggled to discriminate between stimuli presented at these two intervals. Only one-in-four participants met the d' criterion (greater than or equal to 0.75) for classification as a heartbeat detector, a finding replicated by others [52,53].

That only 25% of subjects could discriminate S+ from S– trials was considered an artefact of an excessively stringent criterion and demanding temporal discrimination. Accordingly, several procedural adjustments, such as increasing the delay after each R-wave by 30 ms on S– trials, were made by Katkin *et al.* [54–56], to facilitate discrimination between S+ and S– stimuli. However, these attempts to make the Whitehead task easier were unsuccessful. Furthermore, by making the S+ and S– stimulus trains more easily discriminable [54], the procedural modification

defeated an important feature of the Whitehead task: that S+ and S- stimuli differed only in their temporal relationship to concurrent heartbeats. In the Whitehead task, but not in the Katkin task, the distribution and sequence of interstimulus intervals was identical to the distribution and sequence of inter-heartbeat intervals on both S+ trials and on S- trials (cf. [57,58]). The Katkin task, however, provided non-cardiac temporal cues that could be used to identify S+ and S- trials [59, p. 452]: because the interstimulus intervals were 30 ms longer for S+ signals than S- signals, the time between the first and tenth stimuli was 300 ms longer for the S- than S+ series. Even when some of these methodological concerns were addressed, by using a constant initial delay of R + 300 ms on S- trials, only one-in-five participants discriminated between the S+ and S- stimuli [56]. Davis and co-workers reported that participants could not detect heartbeat sensations on the Davis, Katkin and Whitehead tasks ($A' = 0.55, 0.52$ and 0.53 , respectively). According to Hantas *et al.* [60], only one-in-four participants (17/63) performed better than chance and Katkin *et al.* [55, p. 165], concluded that '...most participants show very little ability to detect their own heartbeats without training'. In sum, none of the aforementioned 2AFC task variants can be recommended.

Not only was there a proliferation of 2AFC variants, but also, as noted by Kleckner *et al.* [61], different versions have employed between 15 and 200 trials. Because classical test theory tells us that precision of measurement increases as a function of the square root of the number of observations, these investigators explored the number of trials required to achieve a sufficiently reliable measure of cardioceptive accuracy without placing excessive burdens on participants. Their analysis suggested that at least 40 trials are required to yield acceptably reliable results and a reasonable estimate of effect size. This important step towards methodological standardization does not, however, explain why only 25% of subjects are identified as heartbeat detectors by 2AFC methods.

A two-interval discrimination task should be easier, the more different are the S+ and S- delays. Stormer *et al.*'s [62] task customized the stimulus delays based on cardiac cycle duration: one quarter (S+) and three quarters (S-) of the previous interbeat interval. In this task, the S+ and S- delays will be more than 256 ms apart (i.e. the gap in the Whitehead task) for anyone with a heart rate of 116 bpm or slower. Participants judged that heartbeats were simultaneous with tones presented early (hits = 69%) and non-simultaneous with tones presented later (false alarms = 27%) in the cardiac cycle. Performance on this task ($d' = 1.14$) was better than on the Whitehead ($d' = 0.61$) and Katkin ($d' = 0.07$) tasks. The relative ease of this task may arise from two features: greater temporal separation between the S+ and S- delays, and, later presentation in the cardiac cycle of the S+ and S-. The latter explanation is compatible with Okifuji *et al.* [63], who compared two Whitehead task variants: discrimination was superior with S+/S- delays of 250/550 ms than with S+/S- delays of 100/400 ms, although both delay pairs were 300 ms apart. Accordingly, the better performance with the 250/550 ms pairing may be attributed to the S+ stimulus occurring closer to and the S- stimulus occurring further from heartbeat sensations [64].

Although performance on the Stormer task is better than on other Whitehead tasks, it is not free of bias. First, individuals with slow and fast heart rates do not differ in the

temporal locations of their heartbeat sensations [65]. Second, the customized delays make the difficulty of the discrimination dependent upon heart rate: the slower the rate, the easier the discrimination, because the S+/S- gap is bigger and the S+/S- fall closer/further from heartbeat sensations.

The convention of classifying simultaneity judgements as correct only for stimuli presented at the S+ delay and incorrect for stimuli at the S- delay originated with Whitehead *et al.* [51], who labelled R + 128 ms stimuli as *immediate* and R + 384 ms stimuli as *delayed* relative to the heartbeat sensation. Since then, many attempts have been made to define delays that optimize the two-interval task, including: 100/31–500 [54], 0/400 and 100/400 ms [66], 100/330–600 [56], 120/380 [67], 200/500 [68], 250/500 [69], 100/400 ms and 250/550 [63] and 300/50 ms [70]. Although it is difficult to make comparisons because of the proliferation of discrimination indices, the tasks yield similarly low levels of discrimination. Overall, about one-in-three/four people reliably judge heartbeat sensations to be simultaneous with S+ stimuli and non-simultaneous with S- stimuli [71]. This poor heartbeat detection performance may be because only this proportion of the population can detect heartbeat sensations and/or the tasks are insensitive psychophysical instruments. Because multi-interval tasks yield higher detection rates (see §4), it is likely that two-interval tasks are relatively insensitive [52].

The problem originates with the flawed assumption that heartbeat sensations occur in the same temporal location relative to the R-wave in all individuals. Take the case of someone tested on the Whitehead task who feels heartbeat sensations at R + 256 ms, and judges stimuli presented at R + 128 ms (S+) and R + 384 ms (S-) to be simultaneous with heartbeat sensations an equal number of times. It is not possible to determine whether non-discrimination between the S+ and S- stimuli is due to an inability to detect heartbeat sensations, or because the S+ and S- stimuli are equally coincident with their heartbeat sensations. This difficulty in interpreting a lack of discrimination will always arise when a heartbeat detection task uses only two intervals, but it may be avoided by using multiple intervals that span the cardiac cycle.

4. Multi-interval tasks: classical psychophysical methods

Clemens [66] was the first to use classical psychophysical methods to identify the temporal location of heartbeat sensations and thereby to specify the optimal S+ and S- delays. In three studies, he asked the participants to press a button if stimuli presented at various intervals after the R-wave were simultaneous with heartbeat sensations. First, he found that signals at R + 0 and R + 100 ms could both be discriminated from signals at R + 400 ms but not from one another. This finding was extended by a further study that used a method of adjustment task to show that participants could not discriminate between signals presented at R + 0, R + 60, R + 120, R + 180 and R + 240 ms, i.e. during the first 240 ms of the cardiac cycle. Finally, using a five-interval method of constant stimuli (MCS) task, Clemens found that signals at R + 0, R + 100 and R + 200 ms were chosen to be simultaneous with heartbeat sensations significantly more than signals at R + 300 and R + 400 ms.

In a six-interval MCS task developed by Yates *et al.* [72], participants judged whether a single light flash presented at $R + 0$, $R + 100$, $R + 200$, $R + 300$, $R + 400$ and $R + 500$ ms was simultaneous with their heartbeat sensations. Heartbeat sensations were more likely to be judged simultaneous with lights delivered in the middle (200, 300, 400 ms) than early (0, 100 ms) phase of the cardiac cycle, thereby challenging the assumption that heartbeat sensations dissipate before $R + 384$ ms [51]. That only eight (40%) participants exhibited modal preferences for the 0, 100 or 200 ms intervals demonstrated that individuals differ in the temporal locations of their heartbeat sensations. Surprisingly, only five (25%) participants met Yates *et al.*'s [72] response consistency and specificity criteria to be classified as heartbeat detectors. A later study by Brener *et al.* [73] indicated that this poor performance was due to data limitations, with a single stimulus presentation being insufficient to make reliable simultaneity judgements.

The potential weaknesses of the previous tasks were corrected by Brener & Kluitse [74], who developed a task based on the method of adjustment and MCS. By pressing one of six buttons, participants could engage an unlimited number of tones at one of six intervals (0, 100, 200, 300, 400, 500 ms) after the R-wave. Pressing a seventh button registered a choice that tones at the prevailing interval were most simultaneous with heartbeat sensations. Tones served as comparison stimuli, because temporal judgements are more accurate in the auditory modality than visual modality [75]. Before engaging in the heartbeat detection task, participants judged light-tone simultaneity to familiarize themselves with the general task demands. The overall distribution of simultaneous choices defines a 'sensation envelope' [76] within which heartbeat sensations are judged to be simultaneous most often with tones presented 200 and 300 ms after the R-wave, and simultaneity was more frequent at $R + 100$ and $R + 400$ ms than $R + 0$ and $R + 500$ ms.

Unlike most other tasks, this one yields measures of the temporal locations of heartbeat sensations as well as the accuracy of heartbeat detection. Brener & Kluitse [74] found considerable individual differences in both of these measures. The temporal locations of heartbeat sensations (modal preferred interval) ranged from 100 to 400 ms, and the specificity of discrimination (standard deviation of the preferred interval) ranged from 43 to 167 ms. Importantly, both performance measures were valid [77] and reliable [65,78]. Independent replications [64,79–82] confirmed that heartbeat sensations were perceived 100–400 ms after the R-wave.

Brener *et al.* explored a simpler task [52] which, like the Yates method [72], was based on the MCS. Participants judged whether a series of 10 tones presented at $R + 0$, $R + 100$, $R + 200$, $R + 300$, $R + 400$ or $R + 500$ ms were simultaneous with heartbeat sensations. Later, it was found that reducing the number of tones to five on each trial did not impair the reliability of the task [73], but abbreviated the administration time considerably. Tones delivered 100, 200 and 300 ms after the R-wave were more likely to be judged simultaneous with heartbeats than tones that were delayed 0, 400 and 500 ms after the R-wave. The reliability (split-half and test–retest) and validity (convergent and discriminant) of this task have been established [52,73,83].

In agreement with previous multi-interval tasks, the MCS reveals substantial individual differences in the temporal

locations of heartbeat sensations, with median preferred intervals ranging from $R + 152$ to $R + 373$ ms. This range of values overlapped with that ($R + 137$ to $R + 323$ ms) exhibited by the same subjects on the Brener–Kluitse procedure [74]. In the MCS procedure, the precision of heartbeat detection was indexed by a continuous performance measure, the interquartile range (IQR) of the distribution of preferred intervals: smaller IQRs imply a narrower heartbeat 'sensation envelope' [76], or more precise and reliable detection of heartbeats. The minimum and maximum IQRs recorded in the MCS procedure (118 and 362 ms) were somewhat greater than those recorded from the same subjects by the Brener–Kluitse task (64 and 314 ms). However, the correlation between the IQRs on these two 6AFC tasks was high ($r = 0.72$), suggesting that they tap the same perceptual ability.

Studies using classical psychophysical methods find that only a quarter of participants judge heartbeat sensations to be most simultaneous with exteroceptive stimuli presented 100 ms after the R-wave. This could explain why approximately one-in-four people show evidence of heartbeat detection in two-interval tasks that present S+ stimuli around this delay. Individual differences in the timing of heartbeat sensations, a prominent and consistent finding in studies that use multi-interval tasks, could also help explain why some people cannot discriminate between the S+ and S– in two-interval Whitehead-type tasks. Support for this proposal comes from a study by Brener & Kluitse [84].

In the first session, the multi-interval heartbeat detection task [74] was used to determine each individual's preferred interval distribution. In the second session, participants completed one of two versions of a 2AFC task. In the customized group, the S+ delay for each subject was set equal to the most frequently chosen (modal) interval on the first session, and the S– delay was their least frequently chosen interval. In the standard group, the S+ delay was $R + 128$ ms and the S– delay was $R + 384$ ms as proposed by Whitehead *et al.* [51]. Not surprisingly, the performance of participants on the customized task was far superior to their performance on the standard Whitehead procedure. This important finding emphasizes a key advantage of multi-interval tasks over two-interval tasks for assessing heartbeat detection. It is not possible to predict accurately when, during the cardiac cycle, individuals sense their heartbeats. Therefore, it is preferable to derive evidence of heartbeat detection using procedures, such as the MCS, that permit individuals to indicate when their heartbeat sensations occur.

While this point has not been disputed, Wiens & Palmer [85] have challenged the use of the χ^2 test to assess the distribution of interval choices generated by 6AFC methods, such as the MCS. On the basis of the group results of standard implementations of 6AFC methods, these investigators argue that individuals who can perceive heartbeats will show a '∩-shaped quadratic trend across intervals'. They also justify the use of a quadratic trend test to qualify subjects as heartbeat detectors on the grounds that multi-modal interval choice distributions are 'completely inconsistent with common theorizing about the utility of the MCS task'. However, the theorizing to which Wiens & Palmer [60] refer is neither referenced nor apparent in the literature. Nevertheless, they do criticize the χ^2 test used with the MCS procedure for being 'needlessly insensitive' to a quadratic trend in the interval choice distribution. While multi-modal distributions of interval choices are uncommon,

they may arise when, for example, a subject detects heartbeat sensations in one bodily location on some trials and in another location on other trials ([41]; see also [86]).

Yates *et al.* [72, p. 564] found that while 45% of their subjects' interval choice distributions did have quadratic trends, the remaining 55% had either linear or higher-order trends. Furthermore, in analysing the results of 109 subjects who had been run on the MCS procedure, Wiens and Palmer themselves found that of 29 subjects who met either the quadratic or χ^2 criteria for classification as heartbeat detectors, nine (31%) had significant χ^2 values but non-significant quadratic trends and were therefore disqualified as heartbeat detectors. Thus, by their criteria, only 18% of the 109 participants were classified as heartbeat detectors, a percentage that is notably lower than in other studies. Furthermore, six participants who qualified as 'heartbeat detectors' by their criteria exhibited interval choice distributions that did not satisfy the χ^2 criterion, implying that their distributions would occur by chance with a probability greater than 5%.

Using binomial probability criteria, Wiens and Palmer also assessed 2AFC performance for two pairs of intervals ($S+/S- = 200/500$ and $200/0$ ms) that could provide a more efficient means of measuring the accuracy of heartbeat detection. The concurrent validities of these two methods and the MCS quadratic trend criterion were then informally compared to that of the MCS χ^2 criterion by calculating the correlations of each of these indices with 'criterion' variables (affect intensity, gender, age) found in some previous studies to be correlated with performance on heartbeat detection tasks [85]. On the basis of these comparisons, the authors concluded that both 2AFC methods ($0/200$ and $500/200$ ms) 'tended to be more sensitive than χ^2 analysis in detecting relationships with variables previously shown to correlate with heartbeat detection' and that this correlational analysis argued for 'the validity of two-interval tasks as measures of heartbeat detection'.

While the analysis provided by Wiens and Palmer of four distinct measures of heartbeat detection derived from a single dataset is novel and interesting, neither of these conclusions is compelling. Prior experiments using 6AFC methods have found no relationship between heartbeat detection and age, gender or body mass index [52,65,74] and hence the lack of correlation between MCS scores and these criterion variables observed by Wiens and Palmer was predictable. Furthermore, it is not surprising that heartbeat detection scores derived by different methods that are themselves only modestly correlated, will be differentially related to criterion variables that had been selected on the basis of their correlations with yet other tests of heartbeat detection. Thus, these correlations provide a weak case for concluding that 2AFC methods are either valid or preferable to the χ^2 test normally employed as a criterion for heartbeat detection with the MCS.

5. Concordance between methods in the classification of heartbeat detectors

The analysis of Wiens and Palmer revealed that the two 2AFC methods ($0/200$ and $500/200$ ms) they recommend are only modestly correlated in their classifications of subjects as heartbeat detectors ($r = 0.24$). The low percentage of variance shared between these methods (6%) re-emphasizes

the need for common standards in the measurement of interoceptive accuracy [25,61].

In this context, while Garfinkel *et al.* [25] classify both the Schandry counting method and 2AFC methods as objective measures of interoceptive accuracy, they note that the methods are only modestly correlated and infer that 'they are founded on distinct (as well as potentially shared) underlying processes' (p. 66). Schulz *et al.* [87] found no correlation between performance on the Schandry task and two 2AFC tasks ($r_s = 0.08$ and 0.22 , n.s.). Similarly, Phillips *et al.* [39] found no significant correlations between performance on the Schandry task and a Whitehead 2AFC task and concluded that 'that performance on one task is largely unrelated to performance on the other task' (p. 508), a judgement seconded by Kleckner *et al.* [61].

In contrast, Hart *et al.* [88] did find that the Schandry task and a 2AFC ($250/550$ ms) task were correlated ($r = 0.38$). Knoll & Hodapp [89] found good correspondence between the Schandry and 2AFC methods for subjects who were either very good or very poor in detecting heartbeats but little correspondence between the methods for the middle ability range of heartbeat detectors. Nevertheless, recognizing that good scores on the Schandry method could be obtained without processing heartbeat sensations, these investigators concluded that 'Schandry's mental tracking task can be used when it makes no difference whether heartbeat perception ability or the ability to estimate heart rate is being assessed' (p. 222). In contrast to the low correlations between different 2AFC methods and between 2AFC methods and the Schandry method, Brener *et al.* [52] found that performance of subjects on the 6AFC Brener and Kluitse and MCS procedures were relatively highly correlated ($r = 0.72$), implying that they are measuring the same or similar perceptual skills.

6. Conclusion

If the ability to detect heartbeat sensations is to be used in evaluating the psychophysiological effects of variations in interoceptive sensitivity, then methodological standardization seems crucial [62]. In the absence of standardization, different methods for probing the accuracy of heartbeat detection will continue to generate different, and even conflicting, conclusions about the role of interoceptive sensitivity.

The 2AFC methods are biased against individuals who experience heartbeat sensations at delays after the R-wave that differ from the $S+/S-$ delays selected by the experimenter. Such individuals may be falsely classified as non-detectors (false negatives). Nevertheless, it must be acknowledged that individuals who meet conventional statistical criteria for heartbeat detection in 2AFC tests are heartbeat detectors (true positives). However, in the heartbeat-counting method, individuals may achieve high scores and be classified as heartbeat detectors on the basis of accurate beliefs but without processing heartbeat sensations. In other words, this test has a false-positive bias in the classification of heartbeat detectors. The different biases of these methods coupled with the unverifiability of the heartbeat-counting method question the equivalence of the methods implied in their designation by Garfinkel *et al.* [62], as objective behavioural methods for tracking 'interoceptive accuracy'.

In contrast, the multi-interval methods provide unbiased alternatives to the 2AFC and heartbeat-tracking methods. Of the 6AFC methods, the MCS [52] is the most efficient. A standard implementation of this procedure involves 20 trials at each of six R-wave to stimulus intervals ($R + 0$, $R + 100$, $R + 200$, $R + 300$, $R + 400$ and $R + 500$ ms). Assuming five heartbeat–stimulus pairings on each trial [73] and a 5 s intertrial interval, the duration of the entire procedure, including an initial 30-trial exteroceptive familiarization task, is approximately 35 min. The validity of the MCS procedure has not been seriously challenged and like other 6AFC methods, it has been shown to be more reliable (split-half, test–retest) than other methods. Rather than prejudging when heartbeat sensations occur during the cardiac cycle, 6AFC methods allow each individual to indicate when during the cardiac cycle his/her sensations occur. The analysis infers that the distribution of interval choices will deviate from chance only if the individual is detecting heartbeat sensations. The distribution of interval choices also provides continuous measures of precision (the interquartile range of the distribution; IQR) and the temporal location of the heartbeat sensation (the median interval; MI).

Individual differences in these measures are correlated with general sensory and perceptual acuity, and sensitivity of somatosensory mechanoreceptors [90]. The measures also show that contrary to common assumptions, neither hypochondriacs [15] nor ‘somatosensory amplifiers’ [16] have elevated interoceptive sensitivity. Furthermore, heart transplant patients, who have undergone cardiac denervation,

are not worse at detecting heartbeat sensations than controls with normal cardiac innervation [79]. This observation reinforces the view expressed earlier that sensing visceral activity may involve not only interoceptors, but also the somatosensory mechanoreceptors and perhaps even the exteroceptors [91, p. 240/241].

Although multi-interval methods, such as MCS, were developed to correct apparent weaknesses in the counting and 2AFC methods, they have seldom been used to investigate the affective and cognitive correlates of individual differences in interoceptive accuracy. In comparison, while the counting method may erroneously classify individuals as sensitive to heartbeat stimuli and the 2AFC methods may erroneously classify individuals as insensitive to heartbeat stimuli, both methods have been reported to predict a variety of affective and cognitive characteristics as well as specific electrophysiological and fMRI patterns of activation. These results could be extended and clarified by replicating the experiments from which they were derived using multi-interval tasks such as MCS.

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References

- Razran G. 1961 The observable unconscious and the inferable conscious in current Soviet psychophysiology. *Psychol. Rev.* **68**, 81–147. (doi:10.1037/h0039848)
- Cameron OG. 2001 Interoception: the inside story—a model for psychosomatic processes. *Psychosom. Med.* **63**, 697–710. (doi:10.1097/00006842-200109000-00001)
- Critchley HD, Harrison NA. 2013 Visceral influences on brain and behavior. *Neuron* **77**, 624–638. (doi:10.1016/j.neuron.2013.02.008)
- Gendron M, Barrett LF. 2009 Reconstructing the past: a century of ideas about emotion in psychology. *Emot. Rev.* **1**, 316–339. (doi:10.1177/1754073909338877)
- Barrett LF, Simmons WK. 2015 Interoceptive predictions in the brain. *Nat. Rev. Neurosci.* **16**, 419–429. (doi:10.1038/nrn3950)
- Pollatos O, Schandry R. 2008 Emotional processing and emotional memory are modulated by interoceptive awareness. *Cogn. Emot.* **22**, 272–287. (doi:10.1080/02699930701357535)
- Werner NS, Peres I, Duschek S, Schandry R. 2010 Implicit memory for emotional words is modulated by cardiac perception. *Biol. Psychol.* **85**, 370–376. (doi:10.1016/j.biopsycho.2010.08.008)
- Werner NS, Jung K, Duschek S, Schandry R. 2009 Enhanced cardiac perception is associated with benefits in decision-making. *Psychophysiology* **46**, 1123–1129. (doi:10.1111/j.1469-8986.2009.00855.x)
- van Elk M, Lenggenhager B, Heydrich L, Blanke O. 2014 Suppression of the auditory N1-component for heartbeat-related sounds reflects interoceptive predictive coding. *Biol. Psychol.* **99**, 172–182. (doi:10.1016/j.biopsycho.2014.03.004)
- Seth AK. 2013 Interoceptive inference, emotion, and the embodied self. *Trends Cogn. Sci.* **17**, 565–573. (doi:10.1016/j.tics.2013.09.007)
- Ainley V, Brass M, Tsakiris M. 2014 Heartfelt imitation: high interoceptive awareness is linked to greater automatic imitation. *Neuropsychologia* **60**, 21–28. (doi:10.1016/j.neuropsychologia.2014.05.010)
- Ainley V, Maister L, Tsakiris M. 2015 Heartfelt empathy? No association between interoceptive awareness, questionnaire measures of empathy, reading the mind in the eyes task or the director task. *Front. Psychol.* **6**, 554. (doi:10.3389/fpsyg.2015.00554)
- Sedeno L *et al.* 2014 How do you feel when you can't feel your body? Interoception, functional connectivity and emotional processing in depersonalization-derealization disorder. *PLoS ONE* **9**, e98769. (doi:10.1371/journal.pone.0098769)
- Mussgay L, Klinkenberg N, Ruddel H. 1999 Heart beat perception in patients with depressive, somatoform, and personality disorders. *J. Psychophysiol.* **13**, 27–36. (doi:10.1027/0269-8803.13.1.27)
- Barsky AJ, Brener J, Coeytaux RR, Cleary PD. 1995 Accurate awareness of heartbeat in hypochondriacal and non-hypochondriacal patients. *J. Psychosom. Res.* **39**, 489–497. (doi:10.1016/0022-3999(94)00166-3)
- Mailloux J, Brener J. 2002 Somatosensory amplification and its relationship to heartbeat detection ability. *Psychosom. Med.* **64**, 353–357. (doi:10.1097/00006842-200203000-00020)
- Critchley HD, Wiens S, Rotshtein P, Ohman A, Dolan RJ. 2004 Neural systems supporting interoceptive awareness. *Nat. Neurosci.* **7**, 189–195. (doi:10.1038/nrn1176)
- Craig AD. 2009 How do you feel—now? The anterior insula and human awareness. *Nat. Rev. Neurosci.* **10**, 59–70. (doi:10.1038/nrn2555)
- Khalsa SS, Rudrauf D, Feinstein JS, Tranel D. 2009 The pathways of interoceptive awareness. *Nat. Neurosci.* **12**, 1494–1496. (doi:10.1038/nrn.2411)
- Pollatos O, Schandry R, Auer DP, Kaufmann C. 2007 Brain structures mediating cardiovascular arousal and interoceptive awareness. *Brain Res.* **1141**, 178–187. (doi:10.1016/j.brainres.2007.01.026)

21. Couto B *et al.* 2015 Disentangling interoception: insights from focal strokes affecting the perception of external and internal milieus. *Front. Psychol.* **6**, 503. (doi:10.3389/fpsyg.2015.00503)
22. Zaki J, Davis JL, Ochsner KN. 2012 Overlapping activity in anterior insula during interoception and emotional experience. *Neuroimage* **62**, 493–499. (doi:10.1016/j.neuroimage.2012.05.012)
23. Gescheider GA. 1997 *Psychophysics: the fundamentals*, 3rd edn. Mahwah, NJ: Lawrence Erlbaum Associates.
24. Cornsweet TN. 1962 The staircase-method in psychophysics. *Am. J. Psychol.* **75**, 485–491. (doi:10.2307/1419876)
25. Stevens SS. 1957 On the psychophysical law. *Psychol. Rev.* **64**, 153–181. (doi:10.1037/h0046162)
26. Garfinkel SN, Seth AK, Barrett AB, Suzuki K, Critchley HD. 2015 Knowing your own heart: distinguishing interoceptive accuracy from interoceptive awareness. *Biol. Psychol.* **104**, 65–74. (doi:10.1016/j.biopsycho.2014.11.004)
27. Bykov KM. 1957 *The cerebral cortex and the internal organs*. New York, NY: Chemical Publishing.
28. Khalsa SS, Rudrauf D, Sandesara C, Olshansky B, Tranel D. 2009 Bolus isoproterenol infusions provide a reliable method for assessing interoceptive awareness. *Int. J. Psychophysiol.* **72**, 34–45. (doi:10.1016/j.ijpsycho.2008.08.010)
29. Dale A, Anderson D. 1978 Information variables in voluntary control and classical conditioning of heart rate-field dependence and heart-rate perception. *Percept. Motor Skills* **47**, 79–85. (doi:10.2466/pms.1978.47.1.79)
30. Schandry R. 1981 Heart beat perception and emotional experience. *Psychophysiology* **18**, 483–488. (doi:10.1111/j.1469-8986.1981.tb02486.x)
31. Hamano K. 1980 Studies of self-regulation of internal activity: the interoceptive detection and control of cardiac activity via training procedure of cardiac-motor coupling. *Jpn. Psychol. Res.* **22**, 168–177.
32. Kleinman RA. 1970 *The development of voluntary cardiovascular control*. Knoxville, TN: University of Tennessee.
33. McFarland RA. 1975 Heart rate perception and heart rate control. *Psychophysiology* **12**, 402–405. (doi:10.1111/j.1469-8986.1975.tb00011.x)
34. Canales-Johnson A *et al.* 2015 Auditory feedback differentially modulates behavioral and neural markers of objective and subjective performance when tapping to your heartbeat. *Cereb. Cortex* **25**, 4490–4503. (doi:10.1093/cercor/bhv076)
35. Carroll D, Whellock J. 1980 Heart rate perception and voluntary control of heart rate. *Biol. Psychol.* **11**, 169–180. (doi:10.1016/0301-0511(80)90053-8)
36. Gannon L. 1980 Cardiac perception and the voluntary control of heart rate. *Physiol. Psychol.* **8**, 509–514. (doi:10.3758/BF03326485)
37. Schandry R, Specht G. 1981 The influence of psychological and physical stress on the perception of heartbeats. *Psychophysiology* **18**, 154.
38. Ring C, Brener J, Knapp K, Mailoux J. 2015 Effects of heartbeat feedback on beliefs about heart rate and heartbeat counting: a cautionary tale about interoceptive awareness. *Biol. Psychol.* **104**, 193–198. (doi:10.1016/j.biopsycho.2014.12.010)
39. Phillips GC, Jones GE, Rieger EJ, Snell JB. 1999 Effects of the presentation of false heart-rate feedback on the performance of two common heartbeat-detection tasks. *Psychophysiology* **36**, 594–610. (doi:10.1017/S0048577299980071)
40. Ring C, Brener J. 1996 Influence of beliefs about heart rate and actual heart rate on heartbeat counting. *Psychophysiology* **33**, 541–546. (doi:10.1111/j.1469-8986.1996.tb02430.x)
41. Windmann S, Schonecke OW, Frohlig G, Maldener G. 1999 Dissociating beliefs about heart rates and actual heart rates in patients with pacemakers. *Psychophysiology* **36**, 339–342. (doi:10.1017/S0048577299980381)
42. Bestler M, Weitkunat R, Keller W, Schandry R. 1988 Heartbeat perception and body position. *J. Psychophysiol.* **2**, 123.
43. Pauli P, Hartl L, Marquardt C, Stalman H, Strian F. 1991 Heartbeat and arrhythmia perception in diabetic autonomic neuropathy. *Psychol. Med.* **21**, 413–421. (doi:10.1017/S0033291700020523)
44. Ring C, Liu X, Brener J. 1994 Cardiac stimulus intensity and heartbeat detection: effects of tilt-induced changes in stroke volume. *Psychophysiology* **31**, 553–564. (doi:10.1111/j.1469-8986.1994.tb02348.x)
45. Reed SD, Harver A, Katkin ES. 1990 Interoception. In *Principles of psychophysiology: physical, social, and inferential elements*. (ed JCLG Tassinary). Cambridge, UK: Cambridge University Press.
46. Hamano K. 1982 The effect of cardiac detection upon heart rate control: an extension. *Jpn. Psychol. Res.* **24**, 118–123.
47. Ludwick-Rosenthal R, Neufeld RWJ. 1985 Heartbeat interoception: a study of individual differences. *Int. J. Psychophysiol.* **3**, 57–65. (doi:10.1016/0167-8760(85)90020-0)
48. Flynn DM, Clemens WJ. 1988 On the validity of heartbeat tracking tasks. *Psychophysiology* **25**, 92–96. (doi:10.1111/j.1469-8986.1988.tb00965.x)
49. Essau CA, Jamieson JL. 1987 Heart rate perception in the type A personality. *Health Psychol.* **6**, 43–54. (doi:10.1037/0278-6133.6.1.43)
50. Pennebaker JW, Hoover CW. 1984 Visceral perception versus visceral detection: disentangling methods and assumptions. *Biofeedback Self-regul.* **9**, 339–352. (doi:10.1007/BF00998977)
51. Whitehead WE, Drescher VM, Heiman P, Blackwell B. 1977 Relation of heart rate control to heartbeat perception. *Biofeedback Self-regul.* **2**, 371–392. (doi:10.1007/BF00998623)
52. Brener J, Liu X, Ring C. 1993 A method of constant stimuli for examining heartbeat detection - comparison of the Brener-Klavitse and Whitehead methods. *Psychophysiology* **30**, 657–665. (doi:10.1111/j.1469-8986.1993.tb02091.x)
53. Wildman HE, Jones GE. 1982 Consistency of heartbeat discrimination scores on the Whitehead procedure in knowledge-of-results-trained and untrained subjects. *Psychophysiology* **19**, 592.
54. Katkin ES, Blascovich J, Goldband S. 1981 Empirical assessment of visceral self-perception: individual and sex differences in the acquisition of heartbeat discrimination. *J. Pers. Soc. Psychol.* **40**, 1095–1101. (doi:10.1037/0022-3514.40.6.1095)
55. Katkin ES, Blascovich J, Reed SD, Adamec J, Jones J, Taublieb A. 1982 The effect of psychologically induced arousal on accuracy of heartbeat self-perception. *Psychophysiology* **19**, 568. (doi:10.1111/j.1469-8986.1982.tb02538.x)
56. Davis MR, Langer AW, Sutterer JR, Gelling PD, Marlin M. 1986 Relative discriminability of heartbeat-contingent stimuli under three procedures for assessing cardiac perception. *Psychophysiology* **23**, 76–81. (doi:10.1111/j.1469-8986.1986.tb00598.x)
57. Brener J, Jones JM. 1974 Interoceptive discrimination in intact humans. *Physiol. Behav.* **13**, 763–767. (doi:10.1016/0031-9384(74)90259-5)
58. Ross A, Brener J. 1981 Two procedures for training cardiac discrimination. *Psychophysiology* **18**, 62–70. (doi:10.1111/j.1469-8986.1981.tb01544.x)
59. Katkin ES, Reed SD, DeRoo C. 1983 A methodological analysis of three techniques for the assessment of individual differences in heartbeat detection. *Psychophysiology* **20**, 452.
60. Hantas M, Katkin ES, Blascovich J. 1982 Relationship between heartbeat discrimination and subjective experience of affective state. *Psychophysiology* **19**, 563.
61. Kleckner IR, Wormwood JB, Simmons WK, Barrett LF, Quigley KS. 2015 Methodological recommendations for a heartbeat detection-based measure of interoceptive sensitivity. *Psychophysiology* **52**, 1432–1440. (doi:10.1111/psyp.12503)
62. Stormer SW, Heiligtag U, Knoll JF. 1989 Heartbeat detection and knowledge of results: a new method and some theoretical thoughts. *J. Psychophysiol.* **3**, 409–417.
63. Okifuji A, Harver A, Katkin ES, Reed SD. 1988 Heartbeat perception: assessment, symptoms, and anxiety. *Psychophysiology* **25**, 473.
64. Brener J, Ring C. 1995 Sensory and perceptual factors in heartbeat detection. In *Interoception and cardiovascular processes*. (ed DVR Schandry). Heidelberg, Germany: Springer.
65. Ring C, Brener J. 1992 The temporal locations of heartbeat sensations. *Psychophysiology* **29**, 535–545. (doi:10.1111/j.1469-8986.1992.tb02027.x)
66. Clemens WJ. 1984 Temporal arrangement of signals in heartbeat discrimination procedures. *Psychophysiology* **21**, 187–190. (doi:10.1111/j.1469-8986.1984.tb00202.x)
67. Lombardo TW, Epstein LH. 1986 The nicotine paradox: effect of smoking on autonomic discrimination. *Addict. Behav.* **11**, 341–344. (doi:10.1016/0306-4603(86)90061-4)
68. Clemens WJ, Guethlein WG, Katkin ES. 1988 Heart beat detection and ear bias. *Psychophysiology* **25**, 440.
69. Harver A, Katkin ES, Bloch E. 1988 Comparison of heartbeat and respiratory sensation among

- male and female subjects. *Psychophysiology* **26**, S22.
70. Gardner RM, Morrell JA, Watson DN, Dandoval SL. 1990 Cardiac self-perception in obese and normal persons. *Percept. Motor Skills* **70**, 1179–1186.
 71. Ring C. 1993 *The detection of cardiac activity*. Stony Brook, NY: SUNY Stony Brook.
 72. Yates AJ, Jones KE, Marie GV, Hogben JH. 1985 Detection of the heartbeat and events in the cardiac cycle. *Psychophysiology* **22**, 561–567. (doi:10.1111/j.1469-8986.1985.tb01651.x)
 73. Brener J, Ring C, Liu X. 1994 Effects of data limitations on heartbeat detection in the method of constant stimuli. *Psychophysiology* **31**, 309–312. (doi:10.1111/j.1469-8986.1994.tb02219.x)
 74. Brener J, Kluitse C. 1988 Heartbeat detection: judgments of the simultaneity of external stimuli and heartbeats. *Psychophysiology* **25**, 554–561. (doi:10.1111/j.1469-8986.1988.tb01891.x)
 75. Doob LW. 1971 *Patterning of time*. New Haven, CT: Yale University Press.
 76. Efron R. 1959 The duration of the present. *Ann. N.Y. Acad. Sci.* **66**, 713–739.
 77. Ferguson ML, Fernquist SK, Harver A, Katkin ES. 1989 Validation of a 'preferred interval' heartbeat detection task. *Psychophysiology* **26**, S22.
 78. Kluitse C. 1987 *Discrimination of cardiac activity*. Hull, UK: University of Hull.
 79. Barsky AJ, Ahern DK, Brener J, Surman OS, Ring C, Dec GW. 1998 Palpitations and cardiac awareness after heart transplantation. *Psychosom. Med.* **60**, 557–562. (doi:10.1097/00006842-199809000-00007)
 80. Bessette PR, Scully BM, Jones GE. 1991 Test–retest reliability of the Brener–Kluitse HB perception paradigm. *Psychophysiology* **28**, S12.
 81. Dedic J, Dahme B. 1997 An experimental study of heart beat detection in subjects with high versus low somatic symptom index under conditions favoring interoception. *Psychophysiology* **34**(Suppl. 1), S30.
 82. Katkin ES, Cestaro VL, Weitkunat R. 1991 Individual differences in cortical evoked potentials as a function of heartbeat detection ability. *Int. J. Neurosci.* **61**, 269–276. (doi:10.3109/00207459108990745)
 83. Schneider TR, Ring C, Katkin ES. 1998 A test of the validity of the method of constant stimuli as an index of heartbeat detection. *Psychophysiology* **35**, 86–89. (doi:10.1111/1469-8986.3510086)
 84. Brener J, Kluitse C. 1988 Temporal location of S+ and S– signals in the Whitehead heartbeat discrimination procedure. *Psychophysiology* **25**, 436–437.
 85. Wiens S, Palmer SN. 2001 Quadratic trend analysis and heartbeat detection. *Biol. Psychol.* **58**, 159–175. (doi:10.1016/S0301-0511(01)00110-7)
 86. Ceunen E, Van Diest I, Vlaeyen JW. 2013 Accuracy and awareness of perception: related, yet distinct (commentary on Herbert *et al.* 2012). *Biol. Psychol.* **92**, 426–427. (doi:10.1016/j.biopsycho.2012.09.012)
 87. Schulz A, Lass-Hennemann J, Sutterlin S, Schachinger H, Vogele C. 2013 Cold pressor stress induces opposite effects on cardioceptive accuracy dependent on assessment paradigm. *Biol. Psychol.* **93**, 167–174. (doi:10.1016/j.biopsycho.2013.01.007)
 88. Hart N, McGowan J, Minati L, Critchley H. 2013 Emotional regulation and bodily sensation: interoceptive awareness is intact in borderline personality disorder. *J. Pers. Disord.* **27**, 506–518. (doi:10.1521/pedi_2012_26_049)
 89. Knoll JF, Hodapp V. 1992 A comparison between two methods for assessing heartbeat perception. *Psychophysiology* **29**, 218–222. (doi:10.1111/j.1469-8986.1992.tb01689.x)
 90. Knapp K, Ring C, Brener J. 1997 Sensitivity to mechanical stimuli and the role of general sensory and perceptual processes in heartbeat detection. *Psychophysiology* **34**, 467–473. (doi:10.1111/j.1469-8986.1997.tb02391.x)
 91. Brener J. 1977 Visceral perception. In *Biofeedback and behavior*, pp. 235–259. New York, NY: Plenum Press.